

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

21st June 2019

**Subject CM2A – Financial Engineering and Loss
Reserving (Paper A)**

Time allowed: 3 Hours 15 Minutes (14.45 – 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *Mark allocations are shown in brackets.*
3. *Attempt all questions, beginning your answer to each question on a separate sheet.*
4. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** A group of employees working in a Private company contribute part of their salary towards a Pension Scheme. The Pension Scheme invests in 3 asset classes, namely, Government securities, corporate bonds and Equities in the proportion 40%, 40% and 20% respectively. Employees have been contributing for the past 5 years.

The returns in each of the financial years are given in the below table:

FY ending March	Government	Corporate bonds	Equities
2014	6.2%	8.0%	5.7%
2015	7.1%	4.2%	-15.0%
2016	6.8%	3.1%	35.1%
2017	6.9%	5.2%	2.25%
2018	6.2%	5.2%	40.0%

Note: Return in each asset class include capital gains/losses and income

- i) Find the Variance of the return of each of the asset classes and the portfolio (Note: Round up to 5 digits after decimal at each step, Use 'n' in the denominator) (6)
- ii) List the key advantages and disadvantages of the following measures of investment risk in the context of equity:
 - Variance of return
 - Shortfall probability
 - Value at Risk
 - Tail Value at Risk
 (6)
- iii) Find the Value at risk for Equities and the portfolio for the next 1 year, if the next 1 year return of Equities and the portfolio follow normal distribution with Mean and Variance as obtained from the Table. Interpret the results. (3)

One set of employees from the group have noticed that Equities have performed very well over the past 3 years on average and therefore want to change the investment pattern towards 100% into Equities. They also know that Country's regulation allows them to do so.

- iv) In the context of behavioural finance,
 - a) State and explain at least 2 biases behind such a decision. (3)
 - b) State and explain equity premium puzzle. (2)

[20]

- Q. 2)** For each statement, state the form of Market hypothesis that the Market exhibits, if the following statements are correct. Explain briefly the reason:

- a) Actively managed Mutual funds can consistently give higher returns than Nifty Index returns due to their expertise in analysing companies' earnings, Balance sheet, etc.
- b) A stock trader has made super normal profit for past 20 years using Bollinger-band technique (a well-known technical tool)
- c) A trust-worthy newspaper reports that XYZ Mining Corp has larger than expected coal reserve which would increase the Market capitalization by 5% and the share price increased by 5%.
- d) A Television reporter traded in a stock and made money on the basis of information in his 'Interview' with the management (before it was broadcast).

[4]

- Q. 3)** An insurance company has just written single premium contracts that require it to make payments to policyholders of Rs.10,00,000 in five years' time. The total single premiums paid by policyholders amounted to Rs.8,00,000.

The insurance company is to invest the premiums in assets that have an uncertain return. The return from these assets in year t , i_t , is independent of the returns in all previous years with a mean value of 5.5% per annum effective and a standard deviation of 4% per annum effective. $(1+i_t)$ is lognormally distributed.

- i) Calculate, deriving all necessary formulae, the mean and standard deviation of the accumulation of the premiums over the five-year period. (8)
 - ii) Calculate the upper and lower quartiles for the accumulated value at the end of 5 years of an initial investment of Rs. 8,00,000. (4)
- [12]

- Q. 4)** You have been provided a three-step recombining risk-neutral binomial tree to value a three-year American Put option with strike 1000 for the price of on a non – dividend paying share with a spot price of 1000. Risk-free interest rates are 5% per annum at all durations, continuously compounded. The option is to be priced using a volatility of 25%.

Price of the underlying security at each node (time in years) are:

Time 0	Time 1	Time 2	Time 3
			2210.3
		1696.8	
	1302.6		1302.6
1000.0		1000.0	
	767.7		767.7
		589.3	
			452.4

At each node of the tree, the risk-neutral probabilities are 0.53 for an upward movement and 0.47 for a downward movement.

- i) By deriving the expectation and standard deviation of the share price at time 1, both from the tree and theoretically, check whether the tree has been calibrated appropriately at Time 1? (3)
*[Hint: The variance of a Lognormal distribution with parameters μ and σ is: $\{ \exp(2\mu + \sigma^2) * \exp(\sigma^2) - 1 \}$]*
 - ii) Use the tree to determine:
 - a) the price of a three-year European Put option with a strike of 1000
 - b) the price of a three-year American Put option with a strike of 1000
 (6)
- [9]

- Q. 5)** A patient has to undergo a Surgery. He decides to use Stochastic dominance to choose between 2 procedures.

We define 'x' as the life expectancy (in integer years) of the patient after a Surgery under 2 procedures namely; procedure A and B.

The life expectancy after the surgery of both the procedures follow Poisson distribution $[\lambda]$ with the Procedure A having a mean of 6 and Procedure B having a mean of 8.

It is also known that Procedure B is the more complicated one with higher risk of death (40% as compared to 30% of A) during the Surgery. The utility function of the patient is such that he wants to live more and also wants to reduce the risk of death during the Surgery.

i) Prove that $u'(x) > 0$ and $u''(x) < 0$, where u is the utility function (1)

ii) Which procedure does the patient choose? (5)

[Hint: Prove the Stochastic dominance for $x=0$ to $x=4$]

[6]

Q. 6) i) Distinguish between IBNR and Outstanding claim reserve. (2)

ii) Explain the assumptions underlying the following methods:

a) Average cost per claim method

b) Bornhuetter-Ferguson method (3)

iii) The number of claims paid each year under a certain cohort of insurance policies are given in the Table below. Calculate the development factor using the basic chain ladder method.

Accident year	Development year			
	0	1	2	3
2011	1,300	1,100	1,600	2,300
2012	1,700	1,600	1,900	2,500
2013	2,200	1,900	2,500	
2014	2,300	1,500		
2015	2,500			

(3)

[8]

Q. 7) Let S_t be a stock price process which follows a geometric Brownian motion according to the following stochastic differential equation (SDE):

$$dS_t = \left(\mu + \frac{1}{2} \sigma^2\right) S_t dt + \sigma S_t dW_t$$

where μ and σ are constants, and W_t is a standard Brownian motion.

Let B_t be a risk-free bond whose price grows deterministically at the risk-free rate r , continuously compounded, according to the formula $B_t = e^{rt}$

i) Derive the SDE for the discounted stock price process $D_t = B_t^{-1} S_t$ (5)

Let $X = f(S_T)$ be a path-independent claim on S_T for some horizon T .

ii) Describe what is meant by a self-financing and replicating strategy for X . (3)

iii) Explain the steps needed to construct a replicating strategy for X that enables the claim to be valued. (6)

[14]

Q. 8) An investor can invest in the following three assets which are uncorrelated with each other.

Asset	Expected return	Standard deviation
A	7%	15%
B	4%	7%
C	3%	0%

- i) Define efficient frontier and calculate the efficient frontier of the investor taking into account the numbers provided in the above table. (8)
- ii) Explain how an investor with a quadratic utility function would select a portfolio from those making up the efficient frontier. (1)

[9]

Q. 9) i) State the SDE under the risk neutral measure for $r(t)$, the default free instantaneous interest rate of interest at time t , under the following two models, defining all notations used:

- a) Hull & White
- b) 2-Factor Vasicek (4)

- ii) Two companies have Zero Coupon defaultable bonds in issue. Both the bonds A and B have 2 Crores and 3 Crores nominal in issue respectively and redeem exactly in 2 years' time. Under a risk neutral measure, each bond defaults at a constant rate. Both the bonds have a recovery rate of 50%.

Evaluate the default rate of two bonds under the following assumptions:

- a) Continuously compounded risk-free rate of interest of 3.5% p.a.
- b) Issue of bond A and B is priced at 1.6 Crores and 2.2 Crores. (5)

[9]

Q. 10) i) Define a Surplus Process, with proper notation. (2)

- ii) Show that the time to first claim follows exponential distribution. (2)

- iii) The number of claims from a portfolio of policies has a Poisson distribution with Parameter 25 per year. The individual claim amount distribution is LogNormal with mean 3 and Variance 1.1. The rate of premium income from the portfolio is 1000 per year.

If the insurer has initial surplus of 1200, estimate the probability that the insurer surplus at time 2 will be negative, by assuming that the aggregate claims distribution is approximately normal.

(5)

[9]
